

**Instructions to the candidates:**

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right side indicate full marks.
4. Assume suitable data if necessary.
5. Use of Scientific Calculator is permitted.

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**Unit III --- Big Data Analytics Life Cycle**

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**Q1) a)** What is the data preparation phase in the Data Analytics Lifecycle? Explain the Analytics Sandbox and the ETLT process in this phase. [9]

**b)** List the different stakeholders of an analytics project. What are the key outputs they expect at the conclusion of the project? [8]

**OR Q2) a)** List the activities to be carried out in the model planning and model building phases. What are the different tools used for these phases? [8]

**b)** What is linear regression? Explain its primary objectives. Differentiate between simple and multiple linear regression. How do you evaluate the performance of a linear regression model? [9]

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**Unit IV --- Predictive Big Data Analytics with Python**

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**Q3) a)** What is logistic regression? How does it differ from linear regression? Explain the sigmoid function and its role in logistic regression. [9]

**b)** Suppose you are given a dataset of emails classified as spam or not spam with features "offer" (1=present, 0=absent) and "free" (1=present, 0=absent). A new email has offer=1, free=1.

| Email | Offer | Free | Spam |
|-------|-------|------|------|
| 1     | 1     | 0    | No   |
| 2     | 0     | 1    | Yes  |
| 3     | 1     | 1    | Yes  |
| 4     | 0     | 1    | No   |
| 5     | 1     | 1    | Yes  |

Calculate the probability that the new email is spam using Naive Bayes. [9]

**OR Q4) a)** How does the Apriori algorithm discover frequent itemsets? Explain the role of support and confidence in association rule mining. [9]

**b)** Explain the process of building a decision tree. What are the criteria used for splitting nodes (Gini index, entropy, information gain)? [9]

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**Unit V --- Big Data Analytics and Model Evaluation**

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**Q5) a)** Perform K-means clustering on the following dataset with K=2. Initial centroids: (2,3) and (8,6). Compute new centroids after each iteration until convergence.

| Point | X | Y | | ----- | --- | --- | --- | | A | 2 | 3 | | B | 4 | 7 | | C | 3 | 5 | | D | 6 | 9 | | E | 8 | 6 | | F | 7 | 8 | | [9]

**b)** How do you handle noise in text data during preprocessing? Explain Bag of Words and TF-IDF in text analytics. [9]

**OR Q6) a)** Explain hierarchical clustering with a suitable example. What are the real-world applications of hierarchical clustering? [9]

**b)** What is the holdout method? Explain the difference between training set, validation set, and test set. [9]

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## **Unit VI --- Data Visualization and Hadoop**

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**Q7) a)** What is a histogram? How is it used to visualize data distribution? Differentiate between a histogram and a density plot. [9]

**b)** What is the Hadoop ecosystem? Explain its primary components (HDFS, YARN, MapReduce). How does MapReduce fit into the ecosystem? [9]

**OR Q8) a)** What is a box plot? Explain its components: median, quartiles, whiskers, IQR, and outliers. [9]

**b)** Explain the role of Apache Pig in Hadoop data processing. What is Apache Spark and how does it complement Hadoop for big data processing? [9]

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## **Examiner Commentary**

This paper mirrors the DS&BDA April 2024 PYQ pattern. Q3(b) tests Naive Bayes with a real calculation. Q5(a) requires iterative K-means computation --- a must-practice problem. Q7/Q8 balance Hadoop theory with visualization concepts. Scientific calculator use is explicitly permitted.